TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Processing of Man-Made Staple Fibre and Their Blends

Running polyester staple — and a polyester/cotton blend formed at the draw frame — through card and draw frame, and learning why man-made fibre cannot simply be treated as clean cotton.

YM-4202 · MODERN YARN MANUFACTURING SESSIONAL · EXPERIMENT 1 · LEVEL 4 TERM 2 · YARN MANUFACTURING SPECIALIZATION

1. Aim and theory

Aim: to process a man-made staple fibre (polyester, with viscose as the common alternative) through the card and draw frame; to form a polyester/cotton blend of a stated ratio at the draw frame; and to identify, from the behaviour of the material itself, the respects in which man-made fibre processing differs from cotton processing — static, lap licking, fibre finish and staple length settings.

1.1 What is different about a man-made staple fibre

A man-made staple fibre is manufactured, not grown. It is extruded, cut to a chosen length, and delivered with every fibre close to the same length and fineness, almost free of trash, but carrying a deliberately applied **fibre finish** — a thin coating that controls fibre-to-fibre and fibre-to-metal friction. Four practical consequences dominate its processing:

- **Uniformity cuts both ways.** With no short fibres and no trash, there is little for the card to remove; card waste falls sharply compared with cotton. But the same uniformity means fibres move together in coherent sheets, and drafting behaviour differs from a natural mixture of lengths.
- **Static electricity.** Polyester is hydrophobic — it absorbs almost no moisture — so the friction of opening, carding and drafting generates static charge that is not conducted away. Charged fibres cling to rollers, repel each other into fly and web disturbances, and make sliver handling difficult. Viscose, which absorbs moisture readily, shows far less static but is weaker, especially wet. Static is controlled by the fibre finish, by adequate relative humidity, and by earthing and anti-static arrangements on the machine.
- **Lap licking.** In the blowroom, the layers of a polyester lap tend to stick to each other — the lap "licks" — and then tear and feed unevenly when the card unrolls it, because of finish, static and the smooth fibre surface. Anti-licking measures (rougher lap surface, adjusted lap-rod pressure, correct

finish level, and in modern lines scutching changes or direct chute feed to the card that avoids a lap altogether) exist precisely because this fault is characteristic of man-made fibre.

- **Longer staple, wider settings.** Man-made staple is commonly cut longer than the cotton it is blended with, and it is uniform in length. Roller settings in drafting zones are fixed by the longest fibres present: a setting correct for cotton will break or strain longer polyester fibres, while a setting widened for polyester will lose control of shorter cotton fibres. Blend processing is therefore always a compromise setting, chosen with the fibre diagram of the mixture in mind.

The **fibre finish** ties these together. Too little finish: high friction, static, fibre damage and roller lapping. Too much: fibres slip uncontrollably in drafting, the lap licks, and drafting waves appear. The finish level supplied by the fibre maker is a processing specification, not a detail.

1.2 Blending, and why it is done at the draw frame

A blend ratio is a ratio **by weight**. Small-scale mixing in the blowroom is hard to hold to a stated percentage, because the components are fed as fluctuating fibre streams. At the draw frame the components arrive as slivers of known weight per unit length, fed side by side at the same speed from separate cans. The blend is then fixed by arithmetic: the share of each component is its total fed weight per unit length divided by the combined fed weight per unit length.

$$\text{Blend \% of a component} = (\text{number of its slivers} \times \text{its sliver weight per unit length}) \div (\text{sum over all slivers of weight per unit length}) \times 100$$

Draw-frame blending also gives the most intimate mixing available in the spinning line short of the card: the slivers are drafted together, fibre by fibre, so the delivered sliver — and therefore the yarn — has the same ratio in every short length, which is what dyeing uniformity later depends on.

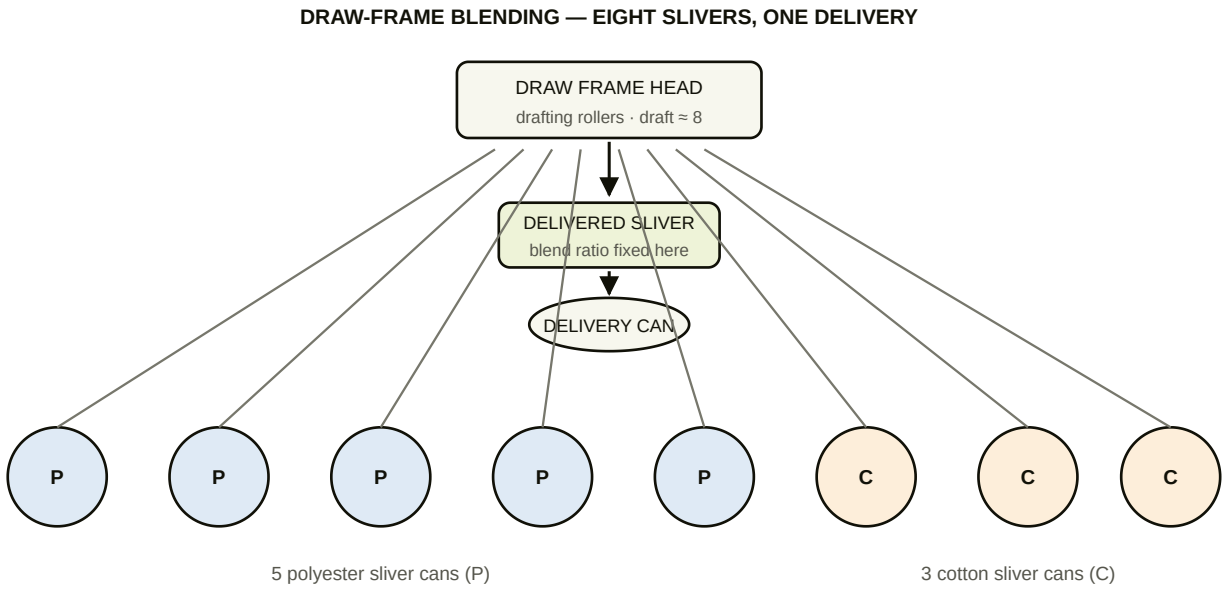


Figure 1 — Draw-frame blending feed. Eight sliver cans creel behind one head; the components are drafted together, so the delivered sliver carries the blend ratio set by the slivers' numbers and weights per unit length.

2. Apparatus and procedure

2.1 Apparatus

ITEM	USE IN THIS EXPERIMENT
Man-made staple fibre	Polyester staple of stated denier and cut length (viscose may be substituted), plus carded cotton sliver for the blend, each lot identified and weighed.
Card	Set for man-made fibre: settings and speeds as directed for the fibre, with attention to web behaviour and static at delivery.
Draw frame	One head creel for eight sliver cans, drafting arrangement with adjustable roller settings, and a delivery can for the blended sliver.
Wrap reel and balance	To establish the weight per unit length (and hank) of each component sliver before blending — the blend calculation is only as good as these two weighings.
Hygrometer	To record relative humidity during the trial; static behaviour cannot be interpreted without it.

2.2 Procedure

1. **Prepare the components separately.** Card the polyester on its own settings and the cotton on cotton settings, so that each component sliver is made under the conditions that suit it. Blending unlike fibres begins only after each has been properly prepared.

2. **Measure each sliver.** From each component, wind a measured length on the wrap reel, weigh it, and record the weight per yard and the hank. Repeat until successive readings agree.
3. **Calculate the creel.** For the target blend (here 65/35 polyester/cotton), work out the numbers and weights of slivers needed, using the calculation of Section 3. Do not assume that counting slivers alone sets the ratio — that is true only when the slivers have equal weight per unit length.
4. **Set the draw frame for the longer staple.** Widen the drafting roller settings to suit the longest fibres in the mixture, set the draft approximately equal to the number of doublings so the delivered hank matches the feed hank, and check the top-roller loading.
5. **Creel the cans** in the calculated arrangement — five polyester and three cotton for this experiment — alternating the components across the creel rather than grouping them, so that any positional effect is shared between the components.
6. **Run and observe.** Run the head and watch for the man-made fibre signatures: web sagging or splitting at the card, fibres clinging to rollers, sliver ballooning or repelling at the creel (static), and any lapping on the drafting rollers. Record the relative humidity alongside every observation.
7. **Test the delivered sliver.** Weigh measured lengths of the blended sliver, confirm its hank, and keep a sample for blend-ratio checking.
8. **Repeat the carding observation** at a changed humidity or finish condition if directed, and compare the static behaviour — the comparison, not a single run, is what teaches the effect.

QUANTITY	POLYESTER SLIVER	COTTON SLIVER	BLENDED SLIVER
Weight per yard (sample data)	3.9 g	3.5 g	3.75 g after draft 8
Slivers in creel	5	3	1 delivered
Blend share found	65.0%	35.0%	100%

The table above carries the **sample data** worked in Section 3; record the trial's own readings in the same form.

3. Worked calculation

WORKED CALCULATION — DRAW-FRAME BLENDING TO 65/35 POLYESTER/COTTON

Case 1 — equal slivers, the common first guess. Eight slivers of equal hank are creeled, five polyester and three cotton. Each sliver then contributes equally by weight:
Polyester share = 5 ÷ 8 × 100 = **62.5%**; cotton share = 3 ÷ 8 × 100 = **37.5%**.
That is a 62.5/37.5 blend, **not** 65/35. Counting slivers sets the ratio only when the slivers weigh the same per unit length.

Case 2 — corrected sliver weights. Keep five polyester and three cotton slivers, but use a polyester sliver weighing **3.9 g per yard** and a cotton sliver weighing **3.5 g per yard** (each established by wrap reel and balance before creeling):
Polyester fed per yard of feed length = 5 × 3.9 = **19.5 g**.
Cotton fed per yard of feed length = 3 × 3.5 = **10.5 g**. Combined feed = 30.0 g.
Polyester share = 19.5 ÷ 30.0 × 100 = **65.0%**; cotton share = 10.5 ÷ 30.0 × 100 = **35.0%** — the target 65/35, met exactly.
Delivered sliver check: with a draft of 8 for 8 doublings, delivered weight per yard = 30.0 ÷ 8 = **3.75 g per yard**, the mean of the feed arrangement — hank is preserved while the ratio is set, exactly as a draw frame should behave.

The lesson generalises: a blend ratio is a statement about **weights**, and sliver numbers are only a convenient handle on weights when hanks are equal. Whenever the component slivers differ — different hanks, or one component deliberately made heavier, as in Case 2 — the calculation must be done in grams per unit length, never in cans.

4. Precautions

- Control static before blaming the machine: maintain the department's relative humidity, confirm the fibre finish is at the maker's specified level, and keep machine earthing intact. Most "mystery" lapping and web faults on polyester are static or finish faults.
- Set drafting rollers for the longest fibre present in the blend. A cotton setting on a longer polyester staple breaks fibres and creates the very short fibre the process is trying to avoid.
- Never creel a blend from unweighed slivers. Weigh each component's sliver first; an assumed hank silently converts a 65/35 order into whatever ratio the slivers happen to give.
- Watch the lap for licking: layers that stick and then tear produce periodic heavy places that no later doubling fully removes. Report licking when seen; do not simply pull the lap apart and continue.
- Keep polyester, viscose and cotton lots and waste strictly identified and separate — a blend ratio is destroyed by careless can handling as surely as by wrong arithmetic.
- Reduce card and draw-frame surface speeds if directed for man-made fibre; heat and friction at high speed aggravate finish softening and roller lapping on polyester.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Six polyester and two cotton slivers of equal hank are creeled at one draw-frame head. What blend ratio results?
2. Why can the 5 + 3 creel of Section 3 give either 62.5/37.5 or 65/35, depending on the slivers used?
3. Give two reasons why polyester generates troublesome static in processing while cotton, under the same conditions, generates much less.
4. What is lap licking, what causes it, and what is its effect on the sliver that follows?
5. Why are draw-frame roller settings for a polyester/cotton blend a compromise, and which component's needs set the minimum setting?
6. Why is blending done at the draw frame rather than by weighing loose fibre in the blowroom?

Answer key

1. With equal hanks each sliver contributes equally by weight: polyester = 6 ÷ 8 × 100 = **75%**, cotton = 2 ÷ 8 × 100 = **25%** — a 75/25 blend.
2. Because the ratio is fixed by fed weight per unit length, not by sliver count. With equal sliver weights the count decides, and 5 of 8 is 62.5%. When the polyester sliver is made heavier per yard (3.9 g against cotton's 3.5 g), the same five cans carry 19.5 g against 10.5 g in every unit of feed, which is exactly 65/35.
3. Polyester is hydrophobic, so it holds almost no moisture to conduct charge away; and its smooth, uniform surface generates charge steadily by friction at every opening and drafting contact. Cotton carries enough absorbed moisture, at normal processing humidity, to leak charge away as fast as it forms.
4. Lap licking is the sticking together of successive layers of a man-made fibre lap, caused by fibre finish, static attraction and the smooth fibre surface. When the card unrolls the lap, stuck layers tear and feed unevenly, producing periodic thick and thin places in the card sliver.
5. The setting must be at least wide enough for the longest fibres, or those fibres are broken or strained between the roller nips — that minimum is set by the longer component, here the polyester. But widening the setting reduces control over the shorter cotton fibres, which can then slip and form drafting waves. No single setting is ideal for both, so the setting chosen is a deliberate compromise informed by the blend's fibre diagram.
6. Because a draw-frame creel is a weighing device in disguise: each sliver's weight per unit length is known and constant, and all slivers are fed at the same speed, so the ratio is fixed by arithmetic and held in every short length of delivered sliver. Loose fibre fed in the blowroom fluctuates from moment to moment, so the ratio wanders along the yarn and shows up later as dyeing variation.
